Economic Prosperity and Happiness

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Motivation

We wanted a project that interested all of us and was related to finance and economics. This way, we can learn something useful, build on it in the future, and use it for resumes or interviews.

Background

- Traditional economic indicators (GDP, income, etc) show a nation's economic performance, but does not necessarily indicate how its citizens perceive their quality of life
- Subjective measures of well-being such as happiness and life satisfaction have been used to obtain a more comprehensive picture of societal welfare
- Income's marginal benefits diminish after a certain point
    - Non-economic factors such as health, social relationships, and institutional trust may play critical roles in shaping overall happiness.

Research Question and Hypotheses

Research Question: How does economic prosperity (GDP per capita) influence national happiness, and what role does health and social support play in this relationship?

1. GDP per capita - positive effect (reduces economic insecurity and satisfies basic needs)
2. Healthy life expectancy - positive effect (increase in longevity)
3. Social Support - strong positive effect (sense of community, trust, etc)

Data Description

Dataset: World Happiness Report (2015 - 2024)

Sample Size: 1,502

Key Variables:

- Happiness Score
- GDP per Capita
- Healthy Life Expectancy
- Social Support

```
Rows and columns: (1489, 4)
       Happiness score      log_gdp  Healthy life expectancy  Social support
count      1489.000000  1489.000000              1489.000000     1489.000000
mean          5.455206     1.690886                66.725319        0.693739
std           1.125604     0.595733                 7.617302        0.210661
min           1.721000    -2.402074                39.000000        0.000000
25%           4.602600     1.484489                62.000000        0.566090
50%           5.476700     1.844431                68.000000        0.738470
75%           6.292800     2.087872                72.000000        0.862290
max           7.842100     2.302585                85.000000        1.000000
```

Data Cleaning and Preparation

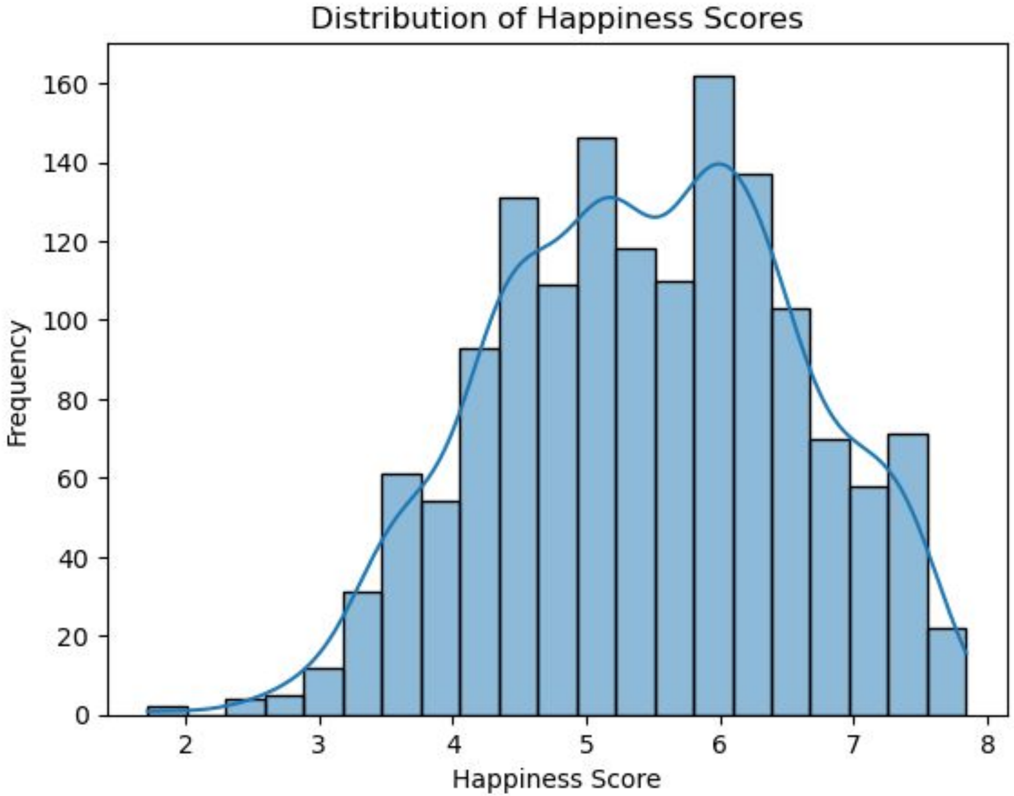
- Removed rows with missing values.
- Converted number formats by replacing commas with decimal points.
- Changed all numeric fields to float so they could be used in calculations.
- Removed infinite values and removed NaNs after transformations.
- Created a variable for logged GDP to reduce skew and improve modeling.

Regression Model and Results

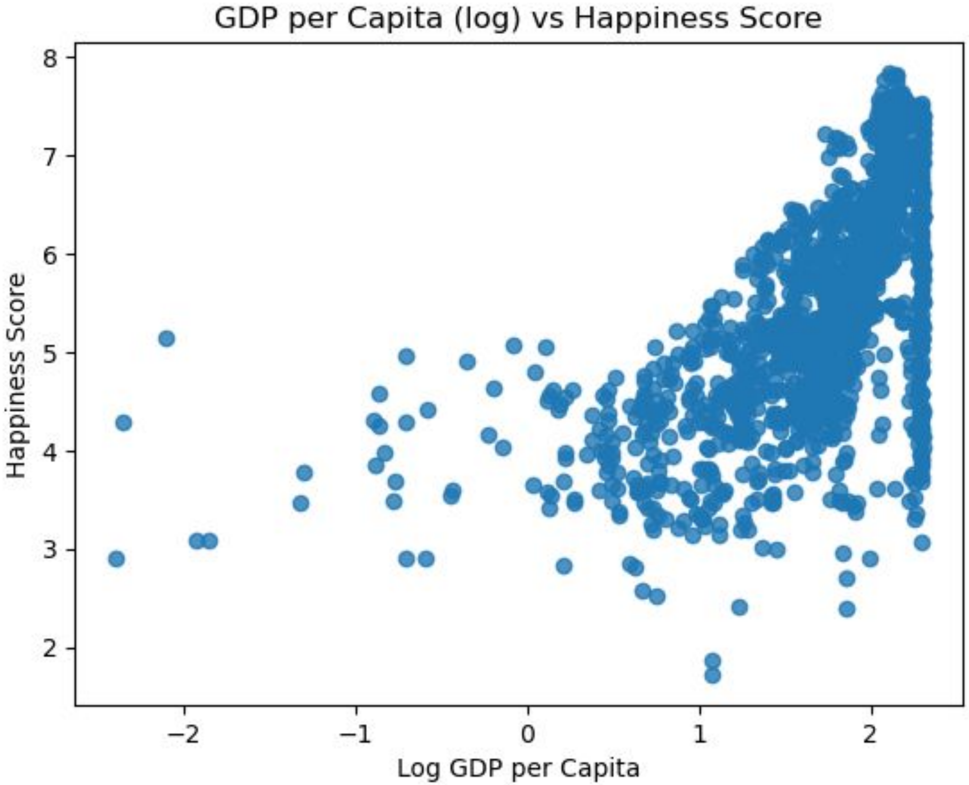
$$HappinessScore_i = \beta_0 + \beta_1 \log GDPperCapita_i + \beta_2 HealthyLifeExpectancy_i + \beta_3 SocialSupport_i + \mu$$

	Coefficient	Std. Error	t-Statistic	P-value
const	0.375	0.1604	2.34	0.0196
GDP per capita	0.0788	0.0092	8.58	0.0
Healthy life expectancy	0.0421	0.0029	14.43	0.0
Social Support	2.5762	0.1018	25.3	0.0

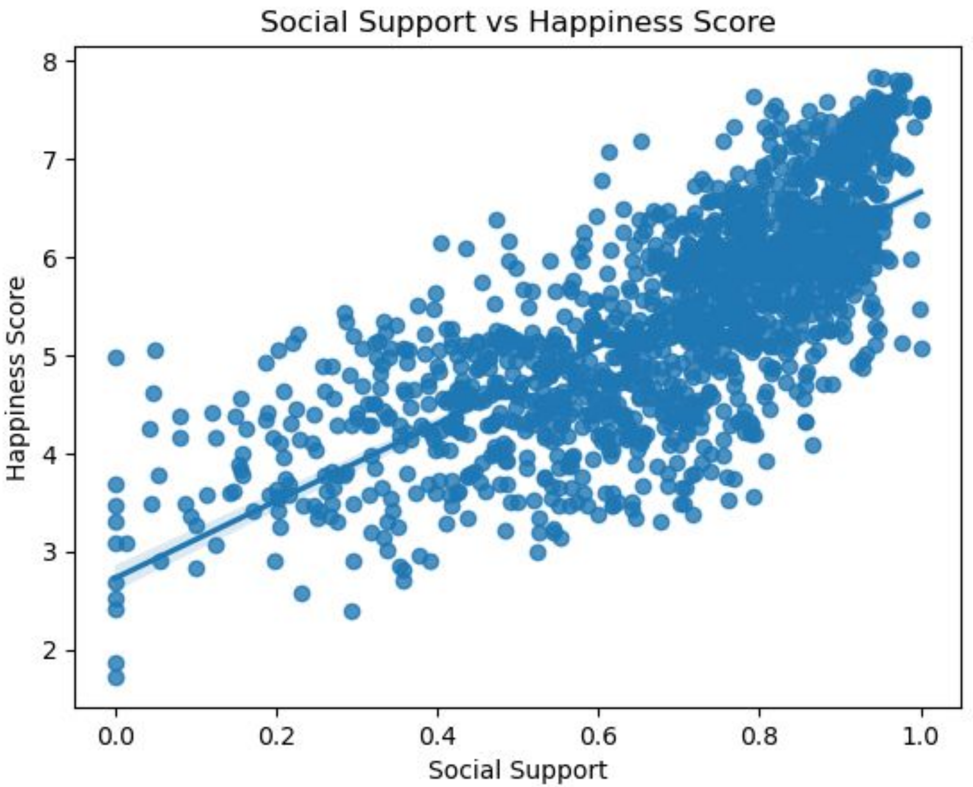
$R^2 = 0.66$



- Approximately bell-shaped with a slight right skew
- Most countries are clustered between a happiness score of 4 - 7



- Positive relationship
- Higher income → Higher happiness score



- Strong, positive, linear relationship
- Countries with higher levels of social support → Higher happiness score
- Social Support had the largest effect

Limitations

- Omitted variable bias
  - Cultural norms
  - Quality of institutions
  - Income inequality
- No causal inference from pooled cross-country model

Future Work and Next Steps

- Build an interactive Tableau dashboard to explore happiness by country and region.
- Create more heatmaps, trend lines, and scatterplots for deeper insights.
- Try advanced models (Random Forest / XGBoost) to find most important factors.
- Overall make the project more visual, dynamic, and useful.

Conclusion

- Economic prosperity is a factor but not the sole factor that explains overall happiness
- Social support → most influential determinant
- Health → significant contributor
- Focus on Economic Growth should not be the only focus for Leaders

THANK YOU

Any Questions?